

Cultivated a strong expertise in data engineering, focusing on the creation of adaptable data solutions that cater to a variety of sectors, utilizing cutting-edge technologies like AWS, Snowflake, and Databricks.

## Core Competencies

Data Engineering Strategy

ETL Pipeline Optimization

Real-Time Data Analytics

Secure Data Infrastructure

Scalable Data Solutions

Business Intelligence Integration

Machine Learning Deployment

End-to-End Data Lifecycle Management

## Education

- Master of Science in Information Systems

University of Maryland, Baltimore County

Maryland, USA

Jan 2023 – Dec 2024

CGPA: 3.93/4.0
- Bachelor of Technology in Electronics & Communication Engineering

SRKR Engineering College, JNTU

India

2018 – 2022

CGPA: 8.23/10.0

## Profile Summary

- Dynamic and results-oriented **Data Engineering Professional** with 3+ years of progressive experience in leading the design, development, and deployment of scalable data infrastructure across diverse industries. Recognized for bridging the gap between business strategy and technical execution through deep expertise in **cloud platforms (AWS, GCP), ETL optimization, and advanced analytics.**
- Proven track record in **leading end-to-end data initiatives**, from pipeline architecture to real-time dashboard delivery, directly influencing **strategic decisions and operational outcomes.**
  - Adept at translating complex business requirements into **robust, secure, and cost-effective** data solutions, leveraging tools like **Snowflake, Databricks, PySpark, and Informatica PowerCenter.**
  - Demonstrated leadership in **mentoring cross-functional teams**, managing stakeholder expectations, and aligning data strategies with organizational KPIs.
  - Strong foundation in **machine learning, NLP, and predictive modeling**, contributing to innovation in real-time personalization and risk analysis.
  - A champion of compliance, and ethical data use, with hands-on implementation of frameworks that ensure regulatory alignment and information security.
  - Recognized for driving **enterprise-level transformation projects**, achieving measurable improvements in scalability, system performance, and analytics readiness.
  - Committed to continuous professional growth with certifications in **AWS Data Engineering, AI Fundamentals, and SQL Development**, positioning as a future leader in cloud-based data architecture.

## Work Experience

Software Engineer  
Igate Solutions  
Texas, USA | Jan 2025 – May 2025

### Leadership Highlights:

- **Directed cross-functional data teams** to build an enterprise-grade analytics pipeline using **AWS Redshift, Glue, and S3**, enabling **real-time data access** and accelerating insights delivery by 40%.
- **Architected and led the development** of modular ETL pipelines with **PySpark & SparkSQL**, optimizing performance and minimizing cost across compute clusters.
- **Championed the transition to Databricks**, introducing a unified platform for data science collaboration and enterprise integration, streamlining development and deployment cycles.
- **Delivered executive-facing dashboards** via **Power BI and Tableau**, turning raw data into strategic KPIs that influenced operational and financial decisions.
- Acted as the **technical liaison** between stakeholders, translating high-level business goals into actionable data strategies and infrastructure blueprints.
- Introduced **governance controls and monitoring frameworks** using AWS CloudWatch and IAM, establishing standards for auditability, performance tracking, and security.
- **Mentored junior engineers**, cultivating a knowledge-sharing culture and ensuring project continuity and quality through internal upskilling and code reviews.

### Key Achievements:

- Reduced end-to-end data latency by **30%**, improving decision speed for key operational processes.
- Led migration of legacy ETL pipelines to cloud-native architecture, achieving **scalability and 25% cost reduction.**
- Recognized by senior management for driving a culture of innovation and **influencing cloud transformation strategy.**
- Co-led a predictive analytics initiative with data science teams, increasing model accuracy, impacting sales and demand forecasting accuracy.

## Certifications

- AWS Certified Data Engineer – Associate
- Oracle Database SQL Certified Associate
- Cisco Certified in Python Essentials
- IBM Certified in AI Fundamentals

## Technical Skills

- **Languages:** Python, R, C, SQL
- **Databases:** MySQL, PL/SQL, Redshift, Snowflake
- **Big Data & Cloud:** AWS (S3, Lambda, Glue, Redshift, EC2, IAM), Databricks, GCP, Kafka, Pub/Sub
- **ETL & Data Engineering:** Informatica PowerCenter, Spark, PySpark, Hadoop, Data Pipelines, Data Warehousing
- **ML & NLP Tools:** Scikit-learn, TensorFlow, PyTorch, TextBlob, Weka
- **Visualization:** Power BI, Tableau, Seaborn, Matplotlib, Plotly, Dash
- **Web Technologies:** HTML, CSS, ReactJS, JSON, JQuery
- **Tools & Frameworks:** Jupyter, Git, GitHub, VS Code, Scrum, Agile

## Research Analyst

University of Maryland, Baltimore County  
Maryland, USA | Jan 2023 – Dec 2024

### Leadership Highlights:

- **Spearheaded an interdisciplinary NLP project**, building a tweet recommendation system with an 85% classification accuracy—shaping strategies in content personalization and digital marketing.
- Drove **strategic research planning** in collaboration with academic and industry partners to align data models with real-world business applications.
- Created scalable visualization assets and automated dashboards that were integrated into **departmental decision-making** and external partner reporting.
- Managed **data quality governance and preprocessing standards** for large datasets (1.6M+ records), ensuring consistent and ethical data use.
- **Mentored undergraduate research assistants**, developing their skills in sentiment analysis, Python, and data interpretation.
- Represented the university at national conferences, showcasing innovations in NLP, predictive modeling, and real-time analytics.

### Key Achievements:

- Delivered a **360° sentiment intelligence platform** to stakeholders in marketing, improving campaign alignment and social media ROI.
- Co-authored a research paper on sentiment-based analytics; cited by peers for application in public policy and brand monitoring.
- Led a dash boarding initiative adopted by 3+ departments for visual storytelling and performance evaluation.

## Programmer Analyst

Cognizant Technology Solutions  
Hyderabad, India | Feb 2022 – Oct 2022

### Leadership Highlights:

- **Led secure data pipeline implementation** for a Fortune 500 banking client, integrating **Informatica PowerCenter** workflows for high-volume data across sensitive financial systems.
- Developed **phishing detection modules** using pattern recognition and behavioral anomaly detection, directly contributing to improved cybersecurity posture.
- **Improved ETL efficiency** by implementing automated validation and data encryption mechanisms, reducing operational risk and manual errors.
- Acted as a key resource in **risk mitigation workshops**, collaborating with cybersecurity teams and business analysts to align data strategies with threat models.

### Key Achievements:

- Achieved a **35% reduction in compliance violations** through strategic pipeline enhancements.
- Recognized with an official **Letter of Appreciation** from Cognizant leadership for exceptional delivery to JP Morgan Chase.
- Contributed to savings in **operational costs** via automation and pipeline consolidation efforts.

## Personal Details

Languages : English, Hindi, & Telugu  
Date of Birth : 22<sup>nd</sup> May 2001

# Key Projects

## 1. Cloud-Based Data Engineering Pipeline for Real-Time Analytics

**Tools/Tech:** AWS (Redshift, S3, Glue), PySpark, Hadoop, SparkSQL, Databricks, Power BI, Tableau

**Role:** Lead Data Engineer | Igate Solutions

- Architected and implemented scalable data pipelines integrating AWS Redshift, S3, and Glue to support high-throughput real-time analytics.
- Led optimization of PySpark-based ETL workflows for large datasets, reducing processing time by 30%.
- Built real-time dashboards with Power BI, Tableau, and Seaborn to deliver actionable insights to business stakeholders.
- Created a pipeline health monitoring framework using AWS CloudWatch.

## 2. Front-End Optimization and API Development for Analytics Web App

**Tools/Tech:** ReactJS, HTML, CSS, JQuery, JSON, REST APIs, Git, CI/CD

**Role:** Full-Stack Developer (Data Engineering Support) | Igate Solutions

- Developed responsive UI for a data-intensive web app, improving user engagement by 30%.
- Designed and integrated RESTful APIs to ensure efficient front-end/back-end communication.
- Achieved 40% faster page loads using React Hooks and reusable components.
- Deployed app via CI/CD pipeline and implemented version control with Git for collaborative development.

## 3. Sentiment-Based Tweet Recommendation System

**Tools/Tech:** Python, NLP, TextBlob, Seaborn, Plotly, Sklearn

**Role:** Research Analyst | University of Maryland, Baltimore County

- Designed a sentiment-aware recommendation engine using NLP, achieving 85% accuracy.
- Processed 1.6M+ tweets using tokenization, lemmatization, and normalization.
- Built visual analytics dashboards with Seaborn and Plotly for presenting sentiment distribution and keyword trends.
- Contributed findings to strategic marketing initiatives and academic research publications.

## 4. Road Accident Prediction and Visualization System

**Tools/Tech:** Python, ML, Clustering, Dash, Plotly

**Role:** Data Science Research Lead | UMBC

- Developed a spatial-temporal model to predict high-risk road accident zones from 145K+ records.
- Applied clustering and anomaly detection for location-based risk modeling.
- Built interactive dashboards with Dash to help public agencies visualize accident hotspots and trends by time and region.
- The project was used in public safety awareness and featured in a research showcase.

## 5. Secure Banking Data Pipeline and Cybersecurity Analytics – JP Morgan Chase

**Tools/Tech:** Informatica PowerCenter, SQL, Data Encryption, Pattern Recognition

**Role:** Programmer Analyst | Cognizant

- Built and secured ETL workflows for financial transactions in a high-compliance banking environment.
- Developed phishing and cybercrime detection modules using pattern analysis of email and log data.
- Participated in enterprise-wide data security audits and contributed to a 35% reduction in threat response time.
- Earned formal recognition from Cognizant leadership for project excellence.

## 6. Sentiment Analysis of E-Commerce Product Reviews

**Tools/Tech:** Python, Logistic Regression, Naive Bayes, SVC, Weka, Tableau, TextBlob

**Role:** Data Analyst

- Analyzed 189K+ product reviews, achieving 88% accuracy using machine learning models.
- Classified sentiments into positive, negative, and neutral to support product performance strategies.
- Visualized insights using word clouds and Tableau dashboards, identifying key sentiment drivers.
- Presented findings to e-commerce stakeholders for marketing and product refinement.

## 7. Task Manager Web App using ReactJS

**Tools/Tech:** ReactJS, UUID, FontAwesome

**Role:** Front-End Developer

- Developed a dynamic task management app using React Hooks and props.
- Incorporated UUIDs for unique task IDs and FontAwesome for a modern UI experience.
- Demonstrated component reusability and modular state management.